MATH840 Final Project

Kristina Polonska

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Final model. A forecast combination of 75% LightGBM and 25% seasonal naive. LightGBM predicts $\log(1+y) - \log(1+\text{level})$, where the level is the mean of the 72 hours ending 48 hours before the forecast hour, using absolute-error loss, hourly and day-level weather features, lagged generation features and a one-year exponential recency weight. The 95% interval comes from backtest errors, scaled by the recent level.

1 What is in the series?

1.1 Generation changes throughout the day

Energy generation changes depending on the hour of the day. So the daily seasonality was found in the data. I discovered this by calculating average generation for each hour and autocorrelation at a lag of 24 hours.

How that affected the model: I included `hour` as a feature and combined LightGBM with 25% seasonal naive to preserve some daily pattern information.

1.2 The generation level changes over time

I compared generation levels across years and examined daily output in October 2026. Generation increased dramatically between 2023 and 2026, but I also observed sudden drops: daily peaks fell from about 2,700 in mid-October to 800–1,000 on 19–25 October, and the daily mean rose again from 258 to 390 on 26 October.

How that affected the model: Instead of predicting the actual generation directly, the model predicts how it differs from a recent generation level. This is done by dividing the actual generation by a level (here it is an average of 72 hourly values). Then a natural log is taken to extract the ratio. The formula as in the code:

$$\ln(1+y) - \ln(1+\text{level})$$

This helps the model transfer pattern across time periods with different levels. In addition, to help the model adapt to level change, there is an exponential recency weighting with the decay period of a year. For every observation is counted:

$$w_t = e^{-\frac{\text{age in hours}}{24\times 365}}$$

1.3 Different weather variables for the day and night

I calculated correlations between daily averages of generation and of the weather, separately for daylight and nighttime hours, in September–October 2026. Daytime generation was positively

correlated with cloud cover ($\approx +0.68$) and negatively correlated with humidity (-0.54). Nighttime generation was positively correlated with wind speed ($+0.69$). Most sudden level changes synchronize with changes in the day's weather.

How that affected the model: I used all seven weather variables and added features such as daily weather averages, daylight averages, the same averages over the three days behind the level and their changes. This allowed LightGBM to capture relationships at both hourly and daily levels and understand how generation might change when the weather changes.

1.4 The model must forecast 48 hours without seeing future generation

Model decision: I used generation lags of at least 48 hours (`lag48`, `lag72`, `lag168`) and rolling averages and maxima calculated from these lagged values. This ensures that, even when predicting the final hour of the forecast, the model only uses historical generation values that were already known when the forecast was made.

2 How did you check the forecast before submitting it?

I used **rolling origin backtesting**. The earlier experiments included 162 reported forecasting windows from September–November 2025 and August–October 2026. For each window, I trained the model using only data available before that date, predicted the following 48 hours, and compared predictions with actual generation and Seasonal Naive.

To better simulate real forecasting conditions, I added artificial errors to historical weather measurements, increasing uncertainty over the 48-hour horizon. I compared different model configurations using RMSE, MAPE, and Winkler score, focusing on hours 25–48 because these determine the private ranking.

Results for hours 25–48

Evaluation	Model	RMSE	MAPE (%)	Winkler	Coverage	Score
Public leaderboard (Day 1)	Seasonal Naive	**115.7**	84.3	521	100%	17.2
	V2 submission	143.2	**55.1**	**426**	88%	14.3
Historical backtest	Seasonal Naive	243.4	50.3	1,525	—	—
	Session 1: 50/50 blend	200.3	43.4	1,150	—	—
	Log target, L1 loss, 75/25 blend	193.8	41.4	1,072	—	—
	Improved model with daily weather and recency weighting	**177.0**	**38.6**	**1,019**	—	—
Local validation (48 hours)	Seasonal Naive	120.40	51.09	568.46	100%	—
	75/25 LightGBM blend	**78.57**	**37.26**	**443.35**	97.9%	—
Local validation (Day 2)	Seasonal Naive	147.68	55.49	654.58	100%	—
	75/25 LightGBM blend	**81.94**	**37.06**	**500.75**	100%	—

The historical backtests showed that adding daily weather features and recency weighting improved the accuracy. The recent validation also showed better results than Seasonal Naive.

3 How good is your interval?

Construction. In the notebook, the model is backtested from every day since 1 September 2026 until the validation window. For every hour I store the error divided by the mean generation of

the 72 hours before the origin, $z = (y - \hat{y})/\text{level}$, because errors grow with the level. The 2.5% and 97.5% quantiles of z are taken separately for four groups: night/day × day 1/day 2. The interval is $\hat{y} + q_{0.025} \cdot \text{level}$ to $\hat{y} + q_{0.975} \cdot \text{level}$, with the lower bound clipped at 0 and both bounds forced to contain the point forecast. The quantiles minimise the Winkler score for $\alpha = 0.05$, and the weather noise in the backtest keeps the interval from being too narrow.

Check. On the 162 windows each window's interval was calibrated on the other 161 windows (leave-one-window-out), so it is never scored on the errors that built it.

Day 2	All 162 windows		October 2026	
	Coverage	Winkler	Coverage	Winkler
SNAIVE interval	87.7%	1525	81.8%	2920
Final interval	**94.7%**	**1019**	**93.7%**	**1241**

The interval is close to the nominal 95% and its Winkler score is about one third lower than the benchmark's. The SNAIVE interval undercovers because its width depends only on the horizon, not on the current level.

4 What is left, and what did not work?

What did not work: I experimented with many forecasting approaches before choosing the final model. Simple methods, including moving averages and SARIMA, performed worse than Seasonal Naive. I also tested different LightGBM objectives, target transformations, blending weights, and feature combinations. Some improved RMSE but performed worse on MAPE or Winkler, making it difficult to find a model that performed well across all three metrics.

What could be improved: Large generation changes remain difficult to predict, and prediction intervals sometimes fail to capture the actual values. The model also performs differently across historical periods, so strong results on one validation window do not guarantee strong results on the competition.

Next steps: I would focus on more reliable prediction intervals, better handling of sudden generation changes, and additional testing of feature combinations and model parameters.